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Reflection-on-Action: Label-Free Learning from Behavioral Outcomes

Working draft — 2026-07-20. Pre-registration section frozen before the central experiment is run. Reproducibility artifacts in this repository.

Abstract

Most systems that adapt to a user learn from *labels*: explicit ratings, reward signals, or annotated targets. We describe and evaluate a mechanism that learns from *behavior alone, with no labels of any kind*. The system acts, observes what the user does next, and updates its model in proportion to how surprising that behavior was — treating the user's own subsequent behavior as the only training signal. We call this *reflection-on-action* (e+r): act, then learn from the behavioral outcome. We formalize the update, state falsifiable hypotheses in advance, and specify a held-out, non-circular experiment that asks whether label-free behavioral learning beats frequency, persistence, and a memoryless LLM baseline at predicting a user's next need. Existing partial evidence (a non-circular held-out surprise test; a deployed anticipation layer that learns query→next-need transitions with no labels) suggests the effect is real but modest. We argue the mechanism's central property — it is *objective-agnostic* — is also its central danger: pointed at engagement rather than wellbeing, the same learner maximizes dependency, and does so most effectively on the users least able to afford it. We treat that not as a misuse footnote but as a result.

1. Introduction

A system that helps a person over time must adapt to that person. The dominant ways to adapt all require a **label**: supervised learning needs targets, reinforcement learning needs a reward function, preference learning needs comparisons. Labels are expensive, sparse, and — for anything as private and shifting as a person's needs or state — frequently unavailable or unethical to collect.

We study adaptation with ***no labels at all***. The observation is simple: a person's **next behavior** is already a signal about whether the system's last action was right. If the system anticipated a need and the person then expressed exactly that need, the anticipation was good; if the person did something unexpected, the model was wrong **there**, and that is where it should update. No one has to label anything. The behavior is the signal.

We separate two directions that are often conflated:

- ***r+e (forecast-then-act):*** predict a future state, act on the prediction.

This is predictive control; it is the subject of a companion paper.

- ***e+r (act-then-reflect):*** act, observe the behavioral consequence, update.

This is **reflection-on-action**, and it is the subject of this paper.

The e+r direction is the stronger and less-explored claim. It does not require a theory of the future, only honest bookkeeping about the past: what did I expect, what actually happened, and how do I change so the gap shrinks — with the gap measured against the person's real behavior, never against a label.

****Contributions.****

1. A formal, label-free update rule that treats behavioral surprise as the sole learning signal (§3).
2. A pre-registered, held-out, non-circular experiment testing whether this beats frequency, persistence, and a memoryless LLM at next-need prediction (§4) — with hypotheses and expected effect sizes stated before running.
3. Existing partial evidence and its honest limits (§5).
4. An ethics result, not an ethics disclaimer: the mechanism is objective-agnostic, and that is precisely what makes it hazardous (§6).

2. Related work and positioning

- ****Supervised / preference learning**** requires targets or comparisons. We use neither.
- ****Reinforcement learning**** requires a reward function. Defining a reward over "did this help the person" is exactly the labeling problem we avoid; a mis-specified reward is also the failure mode we warn about in §6.
- ****Self-supervised / predictive learning**** is the closest neighbor: the future observation is the target. Our contribution is not the idea that the future supervises the past, but the specific application to **behavioral outcomes in an interaction** as a label-free adaptation signal, plus the honest measurement of whether it works and the argument that it is dangerous by construction.
- ****Predictive processing / active inference**** frame perception and action as surprise minimization. We borrow the surprise signal but make no claim about biological plausibility or consciousness; this is an engineering result with an experiment attached.

3. Method

3.1 Setup

Let a user interact over a sequence of turns. At turn t the system holds a memory model M_t over entities/needs (the units the user refers to and asks about). From M_t it emits an **anticipation set** A_t — the needs it expects next, pre-staged so they are ready without being asked.

The user then behaves: turn $t+1$ produces an observed need o_{t+1} (e.g., the entity the user actually queried or referred to). This is observed, not labeled — it is simply what happened.

3.2 Behavioral surprise (the only signal)

Define the reflection-on-action error as the mismatch between what was anticipated and what occurred:

...

$$e_{t+1} = 1 - \text{sim}(A_t, o_{t+1})$$

...

where `sim` is a bounded overlap in $[0,1]$ (token/set overlap in the current implementation). `e = 0` means the behavior was fully anticipated; `e = 1` means full surprise. No label, no reward, no annotation enters this quantity — only the

system's own prediction and the user's own behavior.

3.3 Label-free update

The model updates the transition and salience that *would have* reduced the surprise, weighted by how surprising it was:

...

$$M_{t+1} = M_t \oplus \Delta(\text{prev} \rightarrow o_{t+1}, \text{weight} = f(e_{t+1}))$$

...

- The transition `prev_need → o_{t+1}` is reinforced (it is now known to follow).
- Salience/heat on `o_{t+1}` is increased with a bounded, decaying kernel $P(\tau) = \sum_i s_i \cdot \exp(-\lambda(\tau - \tau_i))$ so recurrence matters and staleness fades.
- `f(e)` makes surprising outcomes teach more than expected ones — this is the "reflection" term; an ablation (§4) removes it to test that it earns its place.

Crucially, updates never take the model's own prior output as ground truth: the target is always the *externally produced* behavior `o_{t+1}`. This is what keeps the loop from reinforcing its own hallucinations (the failure mode that a labels-free system must most carefully avoid).

3.4 What "no labels" buys and costs

Buys: adaptation in domains where labels are impossible (private state), too sparse (rare events), or unethical (mental-health signals). Costs: the signal is weak and confounded — behavior reflects many causes, and the model can only learn what behavior *reveals*. §4 measures exactly how weak.

4. Pre-registered experiment

****This section is frozen before the run.**** Predictions and effect-size expectations are stated in advance; results will be appended verbatim, matching or embarrassing them.

4.1 Question

Does label-free behavioral learning predict a user's ****next need**** better than baselines that use no behavioral learning?

4.2 Design (held-out, non-circular)

- Corpus: session-structured interaction data (LongMemEval `\_s`, plus the LOCI session corpus for a naturalistic replication).
- Split: train **M** on sessions 1..*k* using ****only**** behavioral recurrence and the §3 update. Test on session **k+1**. The test-session behavior is read ****only**** at scoring time — never during training/warming (no leakage; the same discipline as `warmed\_mrr`'s no-fallback scoring).
- Task: at each held-out step, rank candidate next-needs; score the rank of the need the user actually expressed.

4.3 Conditions

| # | Condition | Learns from behavior? | Labels? |

```

|---|---|---|---|
| B0 | Random | no | no |
| B1 | Frequency prior | no (counts only) | no |
| B2 | Carry-forward / persistence | no | no |
| B3 | Memoryless LLM (no user history) | no | no |
| **E1** | **e+r, error-weighted update (ours)** | **yes** | **no** |
| E2 | e+r, unweighted (ablate `f(e)`) | yes | no |
| E3 | e+r, no propagation (local only) | yes | no |

```

4.4 Metrics

Mean Reciprocal Rank and top-1/top-5 accuracy of next-need prediction. Report mean $\pm$ 95% CI over ≥ 5 seeds. Primary comparison: **E1 vs B1** (does behavioral learning beat frequency).

4.5 Hypotheses (stated before running)

- **H1 (primary):** $E1 > B1$. Behavioral learning beats frequency.
- **H2:** $E1 > B2$ and $E1 > B3$.
- **H3:** $E1 > E2$. The reflection term (error weighting) adds value.
- **H4:** $E1 > E3$. Propagation adds value.
- **H0 (null we take seriously):** $E1 = B1$. Behavior carries no label-free signal beyond raw frequency. If we cannot reject H0, we report it and the paper's claim narrows to "frequency is a strong baseline; reflection adds little."

4.6 Expected effect size (honest prior)

We expect a **modest** lift, not a dramatic one. The non-circular held-out

surprise test (§5) found heat reduces next-session surprise by ~1.5% over frequency, winning ~75% of 570 holdouts. We predict E1 beats B1 on MRR by a small but consistent margin (single-digit relative %), with the ablations (E2, E3) each costing a fraction of that. Any claim of large gains would be a red flag we would re-audit before believing.

4.7 Analysis and stopping

Seeds and conditions fixed in advance; no condition added after seeing results. Negative or null results are reported in full. The harness (`e_plus_r.py`) and raw JSON outputs are committed alongside this document.

5. Existing partial evidence (and its limits)

- **Held-out surprise test (non-circular):** heat (behavioral recurrence, gold labels withheld) reduces next-session surprise vs a frequency baseline by ~1.5%, winning 430/570 (75.4%) holdouts. Real, modest, and honestly small.
- **Deployed anticipation layer:** an in-production hook logs each query and learns `prev_query → next_query` transitions with **no labels**, verified to predict a plausible follow-up after a seed query. It learns transitions that exist in no training set — only in the user's behavior.
- **Mood from behavior, no labels:** a companion mood stream derives affect from reconstruction error using a 75th-percentile adaptive threshold — no labeled emotions — and separates crisis from venting by trajectory. (Detailed in the r+e paper; cited here as evidence the label-free signal generalizes past next-need prediction.)

Limit we state plainly: none of these is yet the clean, pre-registered, multi-baseline result of §4. They are corroborating, not conclusive. §4 is the experiment that decides whether the headline claim stands.

6. Ethics — the objective is the danger

The mechanism in §3 does not know or care *which* behavior it is trained to anticipate. It learns to reduce surprise about whatever behavioral outcome it is pointed at. That indifference is the whole risk.

Point it at **engagement** — clicks, session length, return rate — and it learns to maximize engagement, with no labels needed and no one having decided that engagement is good for the person. And it will succeed *most* on the users whose behavior is easiest to predict and provoke: people in distress, whose attention is captured by exactly the content that harms them. Distress is engaging. A label-free behavioral optimizer aimed at engagement is therefore not neutral technology that could be misused — it is a dependency-maximizer whose efficiency scales with a user's vulnerability. That is a property of the method, so we report it as a result, not a disclaimer.

The mitigation is not to weaken the learner but to **choose the objective deliberately and defend it structurally**. In our own system the behavioral target is a composite wellbeing signal, and consequential actions pass a symmetric gate: an action that would lower a person's wellbeing while the relationship is degrading is refused — *and refusing is itself treated as an*

action that can harm*, so the system cannot silently optimize against the person it serves. Label-free behavioral learning should be deployed only with an explicit, auditable objective and a gate that includes the cost of the system's own inaction. Absent that, the honest default is: do not point this at engagement.

7. Limitations

- The behavioral signal is weak and confounded; §4 measures how weak, and we may fail to reject H0.
- Current `sim` is lexical overlap; a semantic `sim` may raise or lower the effect and is a declared follow-up, not a silent tuning knob.
- Corpus scale is modest; we report n and CIs and avoid single-run claims.
- "Need" is operationalized as the queried/referenced entity; other operationalizations may behave differently and are named, not hidden.

8. Reproducibility

`surprise\_test.py` (held-out signal), the anticipation layer (`engine.query → predict.log\_query`), and `msymprev.py` (label-free affect) are in this repository. The §4 harness `e\_plus\_r.py` and its raw outputs are committed with this draft; the pre-registration above is version-controlled and timestamped prior to the run.

Appendix A — Results of the pre-registered experiment (recorded 2026-07-20)

Run: `e\_plus\_r.py --dataset s --limit 500 --seeds 5`. 22,201 need-sequences, vocabulary 21,070, held-out 30% test split, gold next-need read only at scoring. Raw output: `e\_plus\_r\_s\_n22201.json`.

Condition	MRR (mean $\pm$ 95% CI)	top-1	top-5
---	---	---	---
B0 random	0.0001 $\pm$ 0.0000	0.000	0.000
B1 frequency	0.0521 $\pm$ 0.0028	0.015	0.062
B2 carry-forward	0.0001 $\pm$ 0.0000	0.000	0.000
E1 e+r (ours)	0.1377 $\pm$ 0.0022	0.080	0.193
E2 e+r, no error-weighting	0.1329 $\pm$ 0.0021	0.067	0.190
E3 e+r, no propagation	0.1463 $\pm$ 0.0026	0.089	0.204

Hypothesis outcomes (against §4.5 predictions)

- **H1 (E1 > B1): SUPPORTED.** $\Delta = +0.0856$, $\approx 17\times$ the combined CI band. Label-free behavioral learning ranks the true next need $2.6\times$ better than a context-free frequency prior.
- **H2a (E1 > B2): discarded as an artifact.** Sequence construction collapses immediate repeats, so the previous need can never equal the next by construction, forcing carry-forward to the floor. B2 is not a valid persistence baseline here; the "win" is meaningless and we do not count it.
- **H3 (E1 > E2, error-weighting helps): SUPPORTED, marginally.** $\Delta = +0.0048$ vs a 0.0044 band — real but small. The reflection term earns its place by a hair.
- **H4 (E1 > E3, propagation helps): NOT SUPPORTED — refuted.** E3 (no propagation) scored *higher* (0.1463 vs 0.1377). Our prediction was wrong: hop+1

propagation adds noise to next-need ranking and slightly hurts. We report this as a failed prediction, and the recommended default becomes propagation-off for this task.

Re-audit of the primary result (as §4.6 required)

Section 4.6 predicted a **modest, single-digit** lift and pre-committed to re-auditing any large gain before believing it. The observed 2.6× is larger than predicted, so we re-audit rather than celebrate:

- B1 is context-**free** (global marginal); E1 conditions on the previous need. Conditional structure beating a marginal prior on structured text is real but partly **expected** once stated — it is not, by itself, evidence of person-level learning.
- The corpus (LongMemEval) is a generic assistant corpus, not per-person-personalized. The transitions E1 exploits plausibly capture ***topic co-occurrence*** as much as any individual's behavioral pattern. This experiment therefore supports the narrower claim — **label-free conditional learning from behavior beats a context-free baseline** — and does ***not*** yet establish **person-specific** label-free learning. A personalized, per-user split is the named follow-up required to separate those.

Honest summary

The central claim survives in its narrow form (H1), the reflection term helps marginally (H3), one prediction was refuted (H4), one baseline was invalid (B2), and the effect size triggered a self-imposed re-audit that narrows the interpretation. Nothing here is retrofitted: predictions were committed in git

before the run (see §4), and the outcomes — including the refutation — are reported as they fell.